

3.1 String basics ^

Settings

Beta features

string

A string is a sequence of characters that represents textual data like a file location, or a message to the user.

string literal

A string literal is a string defined in the source code of a program with a value with single or double quotes, like 'MARY' or "MARY".

sequence type

Sequence type: a type that orders a collection of objects into a sequence.

PARTICIPATION ACTIVITY

3.1.1: A string is an ordered sequence of characters.

Type a string below to see how a string is a sequence of characters ordered by position. The number indicates the index.

Type a string
(up to 6 characters)

Trish

0	1	2	3	4	5
T	r	i	s	h	

ing the worc

CONSOLE



len()

The `len()` built-in function can be used to find the length of a string

type like a list).

**PARTICIPATION
ACTIVITY**

3.1.4: Using len() to find the length of a string.

Start ☐ 2x speed

```
george_v = "His Majesty George V, by the Grace of God, " \
           "of the United Kingdom of Great Britain and " \
           "Ireland and of the British Dominions beyond " \
           "the Seas, King, Defender of the Faith, Emperor of India"
gandhi = "Mohandas Karamchand Gandhi"
john_f_kennedy = "JFK"

print(len(george_v), "characters is much too long of a name!")
print(len(gandhi), "characters is better...")
print(len(john_f_kennedy), "characters is short enough.")
```

```
185 characters is much too long of a name!
26 characters is better...
3 characters is short enough.
```

Captions ^

1. george_v is a long string literal of almost 200 characters - manually counting each character is tedious.
2. The len() built-in Python function returns the length of any sequence object that is passed to it. len(george_v) counts 185 characters in the string and returns 185. len(gandhi) counts 26 characters in the string and returns 26.
3. john_f_kennedy is assigned with the shortest string literal and len(john_f_kennedy) returns 3.

brackets

A programmer can reference a character at a specific index by appending the desired index after the name of a string variable.

**PARTICIPATION
ACTIVITY**

3.1.6: String indexing.

Start

2x speed

```
alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ "
```

```
print(  
    alphabet[0],  
    alphabet[19],  
    alphabet[19],  
    alphabet[0],  
    alphabet[2],  
    alphabet[10]  
)
```

```
print(  
    alphabet[-14],  
    alphabet[-26],  
    alphabet[-9],  
    alphabet[-8]  
)
```

ATTACK
MARS

Captions ^

1. Individual characters of a string can be referenced using brackets at the end of a string. For example, the first character of alphabet, "A", alphabet[19] is "T", and so on.
2. A negative index starts from the end of the string, which can be useful when a string is very long. For example, alphabet[-1] is the last character of alphabet, or "A".

**PARTICIPATION
ACTIVITY**

3.1.8: Updating a string variable.

Start

2x speed

```
alphabet = "abcdefghijklmnopqrstuvwxyz"
print("Lowercase:", alphabet)

# Change to upper case
alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
print("Uppercase:", alphabet)
```

```
Lowercase: abcdefghijklmnopqrstuvwxyz
Uppercase: ABCDEFGHIJKLMNOPQRSTUVWXYZ
```

Captions ^

1. `alphabet[0] = "A"` tries to update a single character in the string. This is not allowed because attempting to update a character produces a `TypeError`.
2. The correct way to update a string is to assign the variable a new string, as in `alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"`

string concatenation

A program can add new characters to the end of a string in a process called concatenation.

PythonTutor: String concatenation.

```
→ planet = "Mars"
   num_moons = "2"
   moon1 = "Phobos"
   moon2 = "Deimos"

   statement = planet + " has "
   statement = statement + num_moons + " moons."
   statement = statement + " The first moon is " + moon1
   statement = statement + ", and the second moon is " + moon2

   print(statement)
```

→ line that just executed

→ next line to execute

[<< First](#)[< Prev](#)[Next >](#)[Last >>](#)

Step 1 of 9

Program output

Frames

Objects

PARTICIPATION ACTIVITY

3.1.10: Concatenating strings.